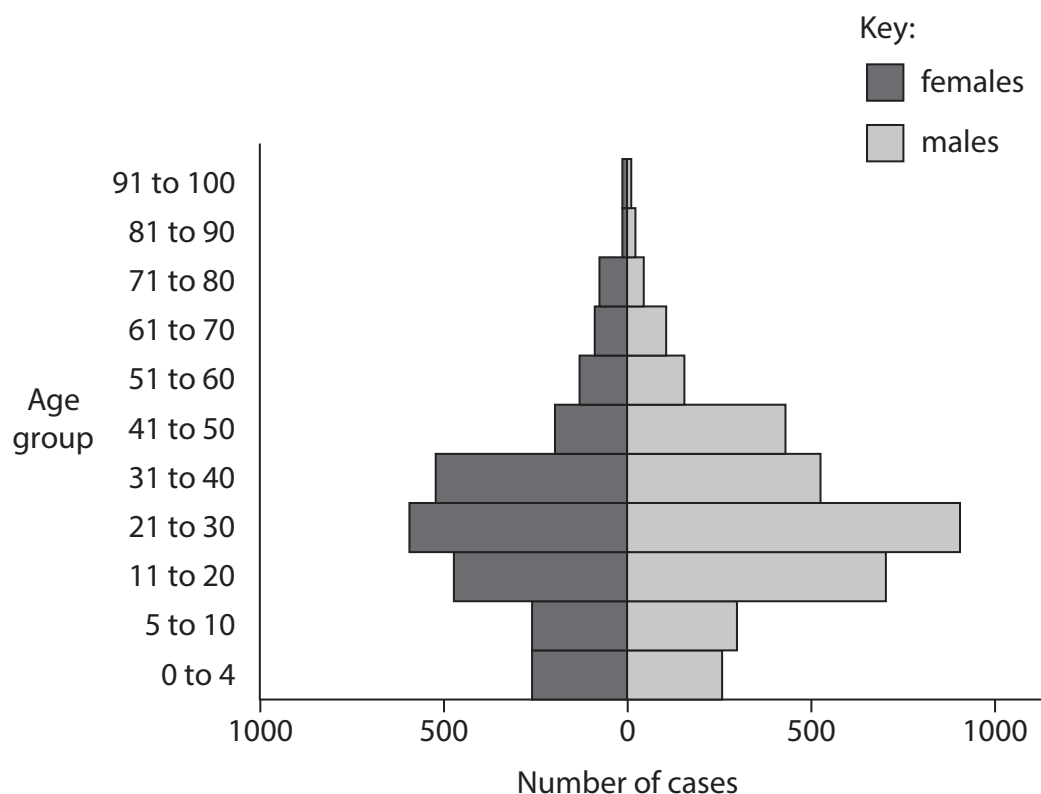


- 8 Since the early 2020s, the number of cholera cases and deaths due to this disease are high as the world is facing its seventh pandemic of cholera.

Cholera is caused by ingestion of food or water contaminated with the bacterium *Vibrio cholerae*.

- (a) The diagram shows the distribution of cholera cases by age group and sex at birth in Malawi, in the first nine months of 2022.



Describe **two** conclusions that can be made about the distribution of cholera cases by age group and sex.

(2)

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(b) *Vibrio cholerae* produce a toxin made up of several subunits.

Genetically modified rice plants are grown that produce rice grains that contain one of these subunits. This subunit acts as a vaccine.

The rice grains were harvested and ground into a powder that can be eaten.

In a clinical trial, three groups of people were given one of three doses of this rice powder vaccine.

Another group of people were given a placebo.

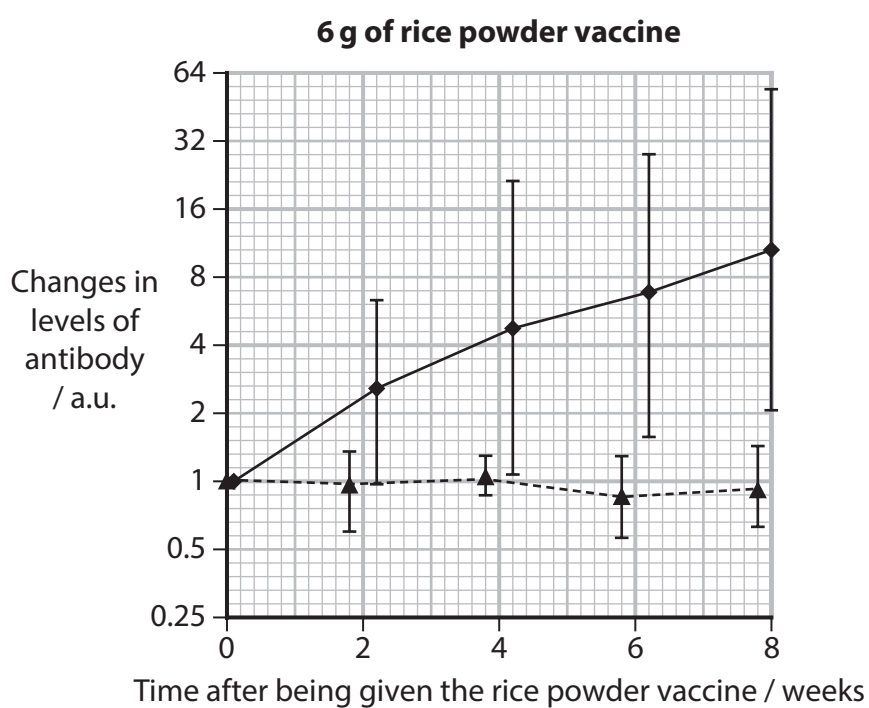
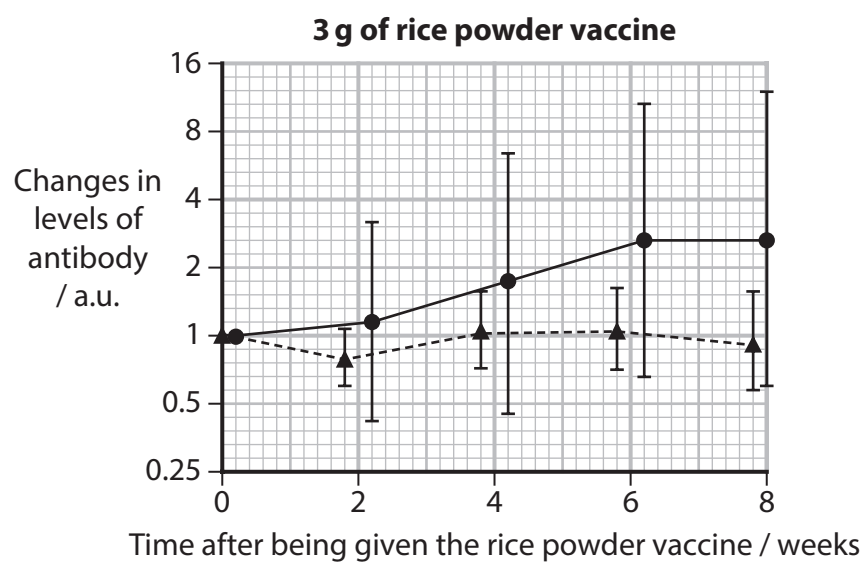
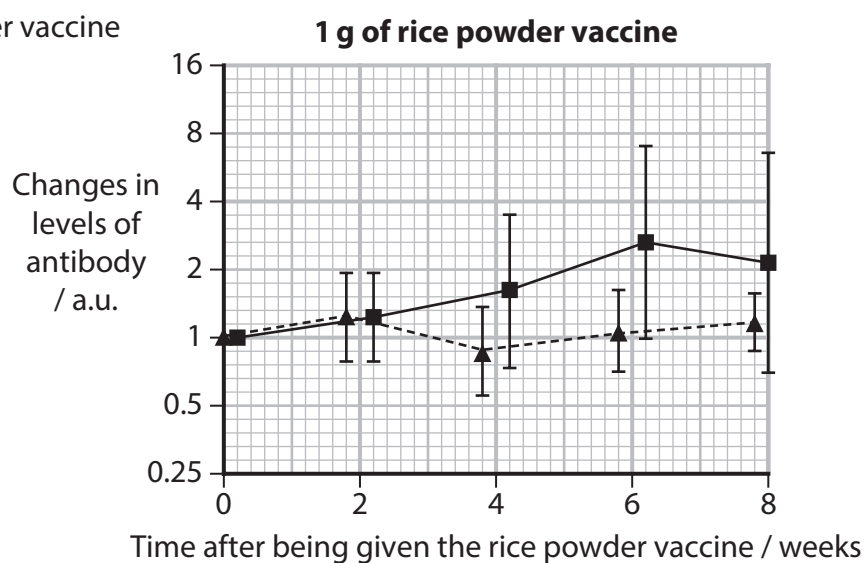
The change in the levels of antibody against the subunit in their blood was measured over an eight-week period.

The graphs show the data obtained.

Key:

— rice powder vaccine

----- placebo



(i) Describe suitable placebos for use in this clinical trial.

Use the information in the question to support your answer.

(2)

(ii) The scientists running the trial concluded that a dose of 6 g of rice powder vaccine was needed to stimulate an immune response.

State why they came to this conclusion.

(1)

(iii) The scientists found that these antibodies also bind to different toxins produced by another type of bacteria.

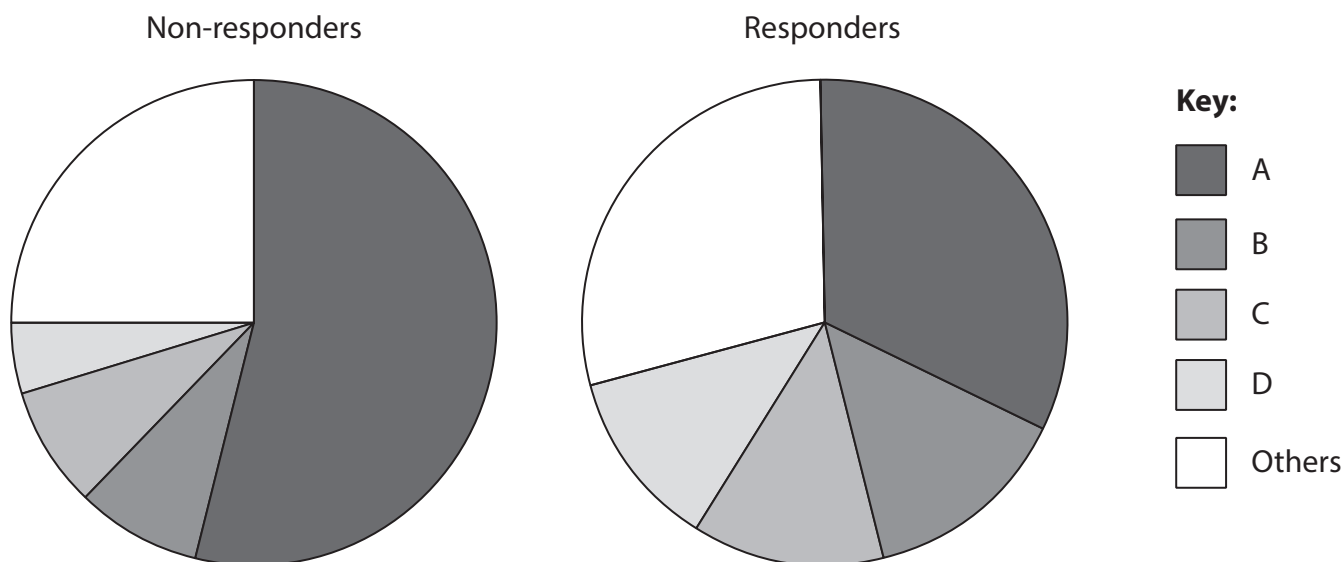
Explain why these antibodies, produced as a result of the rice powder vaccine, could bind to different toxins.

(2)

(iv) Some people in the clinical trial did not respond to the rice powder vaccine.

The scientists determined the proportions of different types of bacteria (A, B, C, D and others) in the gut flora of these non-responders with those of the responders.

The charts show the results.



Compare and contrast the results for the non-responders and the responders.

(3)

[illegible]

- (v) Give **two** limitations when using this data to determine if the type of gut flora could cause a person to be a non-responder to the vaccine.

(2)

(Total for Question 8 = 12 marks)
